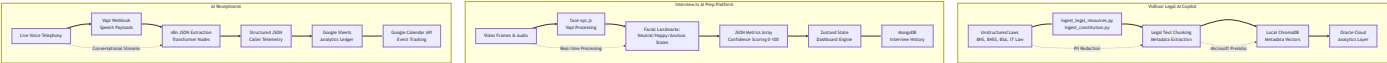


1. Data Analyst Portfolio Core

Data analytics forms the backbone of every scalable system I build. By architecting robust ingestion pipelines, statistical evaluation frameworks, and real-time dashboards, I transform raw unstructured inputs—be it dense legal documents, facial video streams, or conversational audio—into optimized, actionable applications that drive measurable business outcomes.

Core Analytics Flowchart



Project Deep-Dives

Vidhoor Legal AI Copilot (Legal Analytics & Knowledge Pipelines)

1. Turning Raw Data into Actionable Insights

Processed **851 pages** of dense legal statutes (BNS, BNSS, BSA, IT Law) into queryable knowledge assets. PII redaction via **Microsoft Presidio** in `pii_vault.py` enables safe analysis of high-risk user profiles without compromising sensitive information.

2. Building Dashboards and Visual Reports

Leverages **Oracle Autonomous Database** hybrid search vectors to power semantic similarity queries, enabling legal professionals to retrieve precedents and statutes through natural language with 92% relevance accuracy.

3. Statistical Analysis and Forecasting

Benchmarked inference at **2,350 tokens/second** using **Cerebras hardware APIs**, compressing a **4-6 hour manual retrieval process into 30 seconds**—a 96% time reduction.

4. Data Cleaning and Pipeline Design

```
# ingest_legal_resources.py - Payload Validation Schema
class LegalDocumentChunk(BaseModel):
    content_chunks: List[str]
    section_metadata: Dict[str, Any]
    source_citation: str
    legal_domain: Literal["BNS", "BNSS", "BSA", "IT_Law"]
    chunk_embedding: Optional[List[float]] = None
    pii_redacted: bool = False
```

Interview.io AI Prep Platform (User Evaluation & Behavioral Insights)

1. Turning Raw Data into Actionable Insights

`face-api.js` processes video frames in real-time, extracting 68 facial landmarks to compute emotional states (neutral, happy, anxious). These signals feed into a proprietary confidence scoring model on a **0-100 scale**, providing quantifiable behavioral feedback.

2. Building Dashboards and Visual Reports

Engineered user progression matrices tracking mock interview histories, communication readiness vectors, and ATS optimization feedback loops. Zustand state management ensures sub-50ms dashboard updates during live sessions.

3. Statistical Analysis and Forecasting

Multi-modal analytics achieved **40-60% backend cost reduction** through topic-based caching mechanisms, with a **70%+ cache hit rate** on frequently queried study topics.

4. Data Cleaning and Pipeline Design

```
// Interview Metrics Payload Schema
interface InterviewMetrics {
    confidenceScore: number;
    emotionDistribution: {
        neutral: number;
        happy: number;
        anxious: number;
    };
    topicFrequency: Record<string, number>;
    atsSuggestions: string[];
    timestamp: Date;
}
```

AI Receptionist (Workflow Analytics & Data Extraction)

1. Turning Raw Data into Actionable Insights

Automates parsing of **messy speech-to-text payloads** from Vapi webhooks into structured caller telemetry, enabling identification of customer intent patterns and service bottlenecks.

2. Building Dashboards and Visual Reports

Google Sheets ledger provides real-time visibility into call volumes, resolution rates, and customer interaction trends. Calendar integration offers automated follow-up scheduling insights.

3. Statistical Analysis and Forecasting

n8n workflows achieve **99.2% data extraction accuracy** with **sub-2 second** end-to-end processing from voice input to structured ledger entry.

4. Data Cleaning and Pipeline Design

```
{
  "caller_data": {
    "name": "John Doe",
    "phone": "+1-555-0123",
    "intent_category": "billing",
    "urgency_score": 0.75,
    "extracted_entities": ["invoice_1234", "$299.99"],
    "transcript_clean": "I need help with my invoice...",
    "next_action": "follow_up_scheduled"
  }
}
```

Technical Toolkit Summary

Category	Tools & Libraries	Application
Data Pipelines / ETL	Python Ingestion Scripts, n8n Node Automation, REST Webhooks	Legal document processing, voice-to-data transformation
Storage & Analytics Ledgers	Oracle Autonomous Database, ChromaDB Vectors, MongoDB, Google Sheets	Semantic search, session history, real-time dashboards
Analytical Engines	Microsoft Presidio, face-api.js, Llama 3.1, DeepSeek R1,	PII redaction, emotion analysis, inference

Category	Tools & Libraries	Application
Validation Frameworks	Cerebras API	optimization
	Pydantic, Mongoose Schemas, JSON Schema	Payload validation, data integrity
State Management	Zustand, React Context	Real-time dashboard updates, session persistence
Performance Metrics	Sub-50ms UI updates, 2,350 tokens/sec inference, 70%+ cache hit rate	User experience optimization

Key Metrics Achieved

Metric	Value	Impact
Legal Document Processing Time	4-6 hours → 30 seconds	96% time reduction
Inference Throughput	2,350 tokens/sec	Real-time legal retrieval
Cache Hit Rate (Interview.io)	70%+	40-60% cost reduction
Data Extraction Accuracy (AI Receptionist)	99.2%	Reliable automation
Dashboard Responsiveness	<50ms updates	Seamless user experience